

MARVIN — Technical Reference

The exhaustive companion to the [white paper](#): every subsystem, its logic, the decision record behind it, and pointers into the code. Written for contributors and deep evaluators. Covers v0.1.104 (2026-09-03). Where this document and the repository disagree, the repository wins.

Since v0.1.55. Subsystems added after this document's original scope, each with its own decision record: the in-process **pty terminal** (ADR-0078), **background subagents** and built-in read-only agents (ADR-0080), **implementer subagents on isolated git worktrees** (ADR-0081 — the one amendment to the single-assistant rule), **Claude plan usage** from `rate_limit_event` (ADR-0082), the **escalating graph-drift rail** (ADR-0083), **blast-radius and pre-ship impact queries** (ADR-0084/0085), **dependency bootstrap and update checks** (ADR-0086), **Obsidian vault integration** and the plans canvas (ADR-0089 → 0092), the **advisor verdict parser** and its caveat record (ADR-0094/0095, amended 2026-08-31), and **provider-aware model resolution** for OpenRouter (ADR-0096). Two cross-cutting lessons are recorded as ADR-0097 (verify against the surface that *runs*, not the one that *reports*) and ADR-0098 (a rail keyed on vendor tool names is only as durable as those names). Since 2026-09-01: an **LSP client** whose diagnostics come from the buffer (ADR-0099), `/refine` proposing practice lessons (ADR-0101), **two sessions in one worktree** with the collision surfaced (ADR-0102), a **derived implementer-branch lifecycle** (ADR-0103), the **ship-review gate** at `git commit` (ADR-0104, §2.7) and the **practice loop** (ADR-0105, §2.7 and §4).

Paths are relative to the repo root. `runtime/` abbreviates `sidecar/packages/runtime/src/`. ADRs for **this repo** live at `docs/decisions/`; when MARVIN writes ADRs into a **user's project** they go to `<workDir>/docs/adr/`.

Vocabulary (Claude Agent SDK terms used throughout): `canUseTool` — the SDK callback invoked before every tool execution; MARVIN's gate lives there. `agentID` — present on tool calls made by a spawned subagent. `TodoWrite` — the SDK tool the model calls to maintain its task checklist. `Task {subagent_type}` — how a subagent is spawned. `disallowedTools` — SDK-level tool denial for an agent. Graphify terms: edges carry EXTRACTED / INFERRED / AMBIGUOUS confidence tags; "god nodes" are the highest-degree nodes (the de-facto architectural anchors).

1. Core loop & modes

1.1 The single-assistant loop

MARVIN is one Claude session in a continuous user↔assistant loop ([ADR-0001](#)). Workflow "roles" are phases the one assistant moves through, not agents that hand off to each other. This is the founding constraint everything else respects; the 2026 multi-agent literature it is based on reports up to 70% degradation on sequential code work and 17× error amplification in flat agent topologies.

1.2 The 8-phase workflow

Intake → Discovery → Impact analysis → Architecture → Plan → Implement → Verify → Ship, encoded in `runtime/personality.ts` (`CORE_BEHAVIOR`) and explained in [docs/concepts/eight-phase-workflow.md](#). Phases are labelled in chat (`**[Phase N · Name]**`); mutating tools are forbidden before Phase 6; trivial changes take an explicit fast-path. Phase 6 carries a bounded self-remediation contract and Phase 7 a surface-and-offer contract (§11.3).

1.3 Modes: Ask / Agent / Plan

A `mode` axis orthogonal to the permission strategy ([ADR-0036](#)):

Mode	Mechanism	Effect
ask	<code>readOnly</code> flag in <code>classifyToolCall</code> + SDK <code>disallowedTools</code> backstop	every mutating tool hard-denied; reads and graph queries allowed
agent (default)	normal gate	full loop under the auto/gated strategy
plan	read-only planning turn on the advisor/planner model ; plan presented inline; approval runs a separate Agent turn on the executor model	strategy approved before execution; re-planning doesn't arise because execution is not a planning turn

1.4 The plan spine

The plan subsystem is MARVIN's most-revised area; the current design:

- **Plan card** — a plan reply must open `# Plan — <title>`; the native renderer shows it as a collapsible card; detection is content-shaped so it survives transcript replay ([ADR-0036 addendum](#)).
- **Two tiers** — a bare `TodoWrite` checklist renders as a neutral **Task list**; an approved plan renders as a purple **Plan — <title>** strip that persists and ticks off in place (`macos/MARVIN/ToDoListView.swift`).
- **Reconcile, don't clobber** ([ADR-0046](#)) — the active plan owns hierarchical `PlanSteps`; an incoming `TodoWrite` reconciles into them (matched step → status update, unmatched item → nested sub-task) instead of replacing the list. Completion is computed over top-level steps only. Plans live in a revision-aware session list.
- **Join keys + roll-up** ([ADR-0049](#)) — the executor tags every `TodoWrite` item `[N] / [N.M]`, giving a stable join immune to rewording; a step with sub-tasks completes **iff** every sub-task completes (hard invariant, v0.1.50).
- **Persistence** — a presented plan is auto-written to `<workDir>/.marvin/plans/<slug>.md` and opened in the editor; `PlanFile.render` overlays `[x]` on completed steps and appends discovered work, re-persisted on every reconcile. On session load, `replay` reconstructs plan + checklist from the transcript.
- **Continue anchoring** ([ADR-0050](#)) — the Continue control injects the active plan's concrete steps plus a guardrail: resume *only this plan*, no re-auditing.
- **Plan-in-context** ([ADR-0051](#)) — the client sends an authoritative `planContext` snapshot every turn; the runtime appends it as a `<system-reminder>` suffix on the user message (the volatile, uncached tail — prompt-cache-safe, never persisted), so compaction cannot lose the plan.

Code pointers for the spine: the plan model, reconcile logic, `PlanFile` rendering, and `replay` reconstruction live on the app side in `macos/MARVIN/ChatPreviewView.swift`; the two-tier strip and plan card are `macos/MARVIN/ToDoListView.swift` and `PlanCardView.swift`; the per-turn `planContext` injection is in `runtime/sdk-runner.ts`.

1.5 Interactive decisions

When the model reaches a genuine fork it calls `AskUserQuestion`; the gate routes it through the confirm channel (§2.1) and the native `AskQuestionSheet` renders clickable options (single/multi + "Other"), returning the pick as the tool result ([ADR-0040](#)). The sheet registers with **no auto-deny timeout** — a human weighing a decision is never silently overridden (v0.1.40 fix).

1.6 Session history

Turn transcripts hydrate tail-first: the last 200 events paint instantly; "show 200 earlier / show full log" pages the rest in with an "N of M" count ([ADR-0048](#)). The 120MB worst case is reachable but never auto-loaded.

2. Permission & safety

2.1 The structural gate

`canUseTool` (`runtime/sdk-runner.ts`) runs **before** every tool execution. Both permission strategies flow through the same classifier, `classifyToolCall` — there is no second policy hidden in a closure ([ADR-0015](#)):

- **auto** — everything not hard-denied runs; every mutating call is still classified and written to a JSONL audit log (`runtime/auto-audit.ts`, read via `GET /api/audit`). Bypass-only allowances are tagged `auto-mode bypass`: so a later audit can see what gated mode would have prompted on.
- **gated** — three-way classification: auto-allow (reads, whitelisted commands), confirm (Edit/Write/unlisted Bash → a confirm card with the exact diff), hard-deny.
- **Hard-deny floor** — destructive shell patterns (`rm -rf /`, force-push to main, credential-file writes) short-circuit in **both** strategies.

2.2 The subagent read-only invariant

Any tool call carrying an SDK `agentID` that would mutate the workspace is hard-denied in `classifyToolCall` ([ADR-0030](#)). This single choke-point is what makes every subagent pattern (§5) safe; it cannot be configured off. Precision on "would mutate": denial of the file-editing tools (Write/Edit/NotebookEdit) is structural — the tool identity is unambiguous — while mutating *Bash* is the gate classifier's judgment (pattern lists + policy), a distinction worth keeping in mind when evaluating the guarantee. Contrast §2.7's design hooks, which *are* configurable; this invariant and the hard-deny floor are not.

2.3 Change checkpoints & review

The gate snapshots each file's pre-image the first time a session touches it (`runtime/change-checkpoints.ts`, disk-backed under `<dataDir>/checkpoints/`) — [ADR-0034](#). The review window diffs against that snapshot (not git HEAD, which would show the user's own uncommitted work as MARVIN's). Accept advances the baseline; **reject reverse-applies the diff**, restoring the user's uncommitted state — something `git checkout` could not do. Committing clears reviewed-clean files (`reconcileCommitted`). Known v1 blind spot, documented in the ADR: Bash-driven mutations are not pre-imaged.

The review surface is a dedicated resizable window: side-by-side with Split/Inline toggle, single-column rendering for added/deleted files, virtualized rows, and >1500-line diffs gated behind "Show anyway".

2.4 The file-system write channel

User-initiated tree operations go through a second gated channel ([ADR-0008](#)): `fsWritePolicy` (`sidecar/packages/tools/src/fs-write-policy.ts`) classifies seven ops. Delete-to-Trash is auto (reversible); permanent-delete and secret-file writes (`.env*`, `*.pem`, `id_rsa*`) are confirm-danger; writes cap at 5MB. Confirm tokens are session-scoped, one-shot, 60s TTL. Both read and write routes share `fs-sandbox.ts`: canonicalization, symlink + ancestor-symlink escape rejection, path-length and NUL checks. OS→tree uploads require the `X-Marvin-Client` header plus size caps ([ADR-0009](#)).

2.5 The git channel

All git mutations flow through `sidecar/packages/git/` ([ADR-0012](#)): `runGit` is the only place that shells to git (`shell:false`, `GIT_TERMINAL_PROMPT=0`, timeouts, output caps); `argv-guards` whitelist every ref/pathspec/remote/message and reject flag-injection vectors (`-c`, `--exec-path`, `--upload-pack`, ...) as a last line against RCE. `gitWritePolicy` classifies each op: `push --force` to main is always hard-denied; `--force-with-lease` is confirm-danger; dirty-tree branch switches and pulls are denied. Credentials are **inherited, never handled** ([ADR-0013](#)): MARVIN never writes child stdin, never transforms remote URLs, never prompts for git credentials; the user's own helpers (`osxkeychain`, `gh`, `1Password`) answer. Remote stderr is classified into a stable error taxonomy for the UI.

2.6 Background-execution denial

Shell backgrounding (`cmd &`, `nohup`, `setsid`, `disown`) and the SDK's `run_in_background` are gate-denied ([ADR-0038](#), [ADR-0032](#)) — a detached process would finish unreported. The honest alternative is §9.1.

2.7 Runtime design hooks

`runtime/design-hooks.ts` enforces the most load-bearing behavioral rules **structurally**, below the prompt. Four are built in: a blind source-file read on a codebase question is denied until the graph is consulted (`graphify-first`); a stale graph escalates a nudge (`graph-drift`, [ADR-0083](#)); an edit in `security/schema/CI/migration` paths is denied until an advisor consult has happened (`advisor-on-ADR`); and, since v0.1.102, a `git commit` is refused until the review skills the personality requires have run (`checkShipReview`, [ADR-0104](#)). The ship-review gate reads the diff the commit seals — the index plus

whatever the same command stages: a boundary path (auth / creds / CI / sudoers / .env / *.sh / migrations) requires both `pr-review` and `security-audit`, more than three files or fifty lines requires `pr-review`, docs-only and lockfile-only pass; a review this turn covers the turn, an earlier one holds until the next commit; two denies per skill per turn, then allow and log; fails open on git errors.

Practice rules (ADR-0105) are the fifth family and the first driven by data rather than code.

`checkPracticeRules` reads the rules the user accepted in the Practice pane from `~/.marvin/practice/<projectId>/` (mtime-cached) and enforces each at its tier: `prompt` (a line in the system prompt, via `practicePromptBlock`), `nudge` (an advisory line on the tool result) or `deny` (a refusal, offered only for kinds with a machine-checkable discharge). Since v0.1.103 the four built-in gates are themselves rows in that store (`builtin:*`), so their tier, on/off state and message are editable from the pane; when no row exists the gate runs natively, which is why the 93 gate tests did not move. ADR-0104's brakes apply to every family: a deny needs a discharge path, `measure` mode logs instead of denying, two denies per rule per turn, and a bypass is logged. Since v0.1.104 three more MARVIN-owned hooks came out of the first practice report: a bare project-local skill name is rewritten to its plugin-namespaced form at the gate (the ADR-0058 `updatedInput` mechanism), a `PostToolUseFailure` hook remembers a failed Bash command for the turn so an identical re-run gets an advisory nudge, and the turn-close hook blocks, once, a three-plus-edit turn under an open plan that never updated `TodoWrite`.

Under the default `enforce` mode a deny is unconditional; `measure` logs what would have been denied; `off` disables the layer (`MARVIN_DESIGN_HOOKS`). These rules are belt-and-suspenders: `prompt` contract (§3.1, §5, §11) *plus* runtime hook — unlike the §2.2 invariant and the deny floor, which have no off switch. This layer exists because audits showed prompt-only rules decay as prompts grow: the 2026-09-02 session audit that produced the ship-review gate found the review skills invoked 0× across eight pushes of CI, sudoers and credential changes while every mechanical rule in the same session held.

3. Context & knowledge

3.1 Graphify-first

For structural questions the graph is queried **before** files are read — a hard rule with per-tool MUST triggers in `personality.ts`, no judgement call ([docs/concepts/graphify-integration.md](#)).

Measured 27.5× cheaper per structural question (`graphify benchmark`, 2026-08-15; an earlier ~36× figure was an estimate, never measured) than file-reading; answers cite `file:line`.

3.2 Two graphs, three scopes

Per project (ADR-0028): a **code graph** (`graphify-out/graph.json`, AST-extracted) and a **knowledge graph** (`graphify-out/knowledge/graph.json`, heading structure + cross-links of `docs/ADRs/memory`). All six MCP tools accept `scope: "code" | "knowledge" | "all"` (default `"code"`).

3.3 The `marvin-graph` MCP

In-process server (`sidecar/packages/graphify-bridge/`), read-only, blanket-allowed at the gate: `graph_summary`, `graph_search`, `graph_neighbors`, `graph_query`, `graph_path`, `graph_save_result`. Edges carry EXTRACTED / INFERRED / AMBIGUOUS confidence tags; god nodes and communities are first-class.

3.4 Lifecycle & the context budget

While the IDE has a project open, `/api/chat` fires debounced, fire-and-forget AST rebuilds of both graphs each turn — scoped to the active project, never MARVIN's own repo, zero LLM cost ([ADR-0041](#)). The same ADR budgets first-message context: `buildProjectContext` (`sidecar/packages/project-context/`) injects a graph header, the **ADR titles index** (not bodies), the **memory tail** (not the log), curated docs whole, and open backlog items. Measured effect: ~566K → ~13.4K tokens on a mature production project.

3.5 Project fingerprint & infra probes

`fingerprint.ts` derives ~10–42 namespaced tags (`framework:next@16`, `architecture:multi-tenant`, `compliance:gdpr`, `test:playwright`) from manifests and code tells, cached at `.marvin/fingerprint.json` ([ADR-0024](#)); it drives skill enablement and recommendations (§6). Project-agnostic infra probes (`probeHttp`, `probeDockerContainer`) back Phase 2's "probe running infra" without any hardcoded service list. Both files live in `sidecar/packages/project-context/src/`.

4. Cross-session persistence

Content class determines the store; each store has an **enforced write path** that rejects the wrong class:

Class	Store	Write path	ADR
Durable facts (invariants, gotchas, constraints)	<code>.marvin/memory/<slug>.md</code> + one-line index	<code>remember</code> MCP tool (caps, supersede-by-name, rejects activity/status)	0042
Deferred actionable work	<code>.marvin/backlog/<slug>.md</code> + index	<code>backlog_add</code> / <code>backlog_resolve</code> (consent- gated; <code>provisional:true</code> auto-parks at discovery)	0044, 0047
Material decisions	<code><workDir>/docs/adr/NNNN-*.md</code> (user projects; this repo uses <code>docs/decisions/</code>)	Phase 4, enforced template with Scope-of-Done	triggers in §11.4
Status / activity	git history, changelog	ordinary commits	—
Session notes	<code>.marvin/session-notes.md</code>	the native Scope-met chip	0042
Behaviour — repeat failures, the rules about them, whether they held	<code>~/.marvin/practice/<projectId>/</code> (findings ledger, rules, runs; config and score weights shared across projects)	the Practice pane only, over <code>/api/practice: approve /</code> <code>dismiss / fixed / escalate / adopt</code> . The nightly runner writes findings, never rules	0105

On the backlog's two-step capture: `provisional:true` auto-parks a *memo* the instant something is noticed (no go-ahead needed — a memo is not a commitment); the consent gate is the keep/dismiss review at the scope-met handoff. Capture is automatic, retention is consented.

The practice store is the one layer under the data dir rather than the project's `.marvin/`, because it is about MARVIN, not the project. A finding is keyed by a fingerprint (`kind[:qualifier]`) that a deterministic extractor computes over a transcript, counted by *distinct session*, with day-two semantics (new / recurring / proposed / regressed / confirmed / dismissed-then-resurfaced) and a versioned extractor. Twelve kinds ship, nine with a paired success so a finding carries a rate; three are report-only (cache re-creation, repeated errors, over-budget turns) because they are about MARVIN's implementation, not a behaviour a rule can change. Scoring is a small linear model over recurrence, cost, rate, reliability and actionability, minus decay; the weights can be fit from every ledger's own outcomes (Spearman rank correlation, coordinate descent; cost-share ranking below eight labels) and are applied only on a click, with provenance. A project with fewer than three sessions cannot propose; it shows a calibration counter and offers rules confirmed in the user's other projects for adoption, scoped to this project with its own verification clock. Subagent calls are never counted. No model reads a transcript; the one model call in the pane drafts a rule's wording from aggregates, in a read-only two-turn session.

The design was forced by a real failure: a memory file activity-logged to 419KB (~~99% redundant with ADRs/git~~) that overflowed the context window. ~~/memory-compact~~ migrates legacy logs. The backlog is a **parking lot, never a queue** — no agent pulls from it (Golden Rule 1); open items surface in the next session's context and in the macOS backlog panel. Transcripts (`~/.marvin/sessions/*.jsonl`, every event verbatim) and the cost ledger (`~/.marvin/cost-tracker.json`, daily/weekly/lifetime per project) complete the picture.

5. Subagents

Three sanctioned patterns, all read-only by the §2.2 invariant, all bounded; anything else is forbidden by [ADR-0001](#):

- **Advisor** ([ADR-0007](#), [ADR-0033](#)) — a registered agent definition carrying the user's advisor model and its own reasoning effort; spawned via `Task {subagent_type:"advisor"}`. Blunt-critique structure: risks / alternatives / pushback / verdict. Eight MUST triggers (ADR writes, security-sensitive work, blast radius ≥ 5 , non-backward-compatible changes, architecture tie-breaks, concurrency, crypto, user request) plus anti-triggers.
- **Scout** ([ADR-0014](#)) — breadth-first read-only research (`disallowedTools` at the SDK level + the gate invariant); MUST for 3+ parallel searches or context relief; MUST NOT be sequential-implementation or user-facing authority.
- **Dynamic workflows** ([ADR-0030](#)) — script-orchestrated parallel read-only fan-outs for audits / surveys / migration discovery; opt-in (high effort or explicit ask), never automatic, never implementation.

A standing pattern rather than a type: after drafting an ADR, an advisor "future-MARVIN critique pass" lists every question the ADR leaves unanswered; gaps are closed before the user sees it.

6. Skills

- **Installed $\neq$ active** ([ADR-0037](#)) — `skill-enablement.ts` computes the active set: explicit user choice, else core skills $\cup$ fingerprint-matched domain skills. The prompt names the active set each turn and tells the model to ignore the rest (measured 20 $\rightarrow$ 7 relevant on MARVIN's own repo). Per-skill toggles in the Skills pane persist to `.marvin/skills.json`.
 - **Deterministic triggers** — each core skill (test-driven-development, systematic-debugging, pre-review, security-audit, frontend-design, graphify) has enumerated MUST / MUST-NOT trigger lists in `personality.ts` with no judgement escape hatch in the wording. To be precise about the guarantee: these are prompt-level contracts — enumerated form measurably outperforms soft language, but only the runtime layers (§2) are deterministic. The redesign followed a 2026-05-22 transcript audit that found five of six skills had fired approximately zero times across thousands of qualifying turns.
 - **Skill audit** ([ADR-0024](#), [ADR-0025](#)) — a `## Skill audit pending` block injects until `.marvin/skills.md` exists; MARVIN owes exactly one chip-strip recommendation with two verbs never mixed: **install** (user-global, `~/ .claude/skills/`) for language/framework/test tags, **build** (project-local, `.marvin/skills/`, PR-reviewable, via skill-creator's eval loop) for architecture/domain/compliance tags.
 - **Fetch from Git** ([ADR-0039](#)) — "Add from GitHub" shallow-clones a repo / sub-path / plugin marketplace, discovers `SKILL.md` folders, installs; clone-and-copy only, never executes repository code.
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7. Models & auth

- **Role routing** ([ADR-0002](#), [ADR-0003](#)) — independent executor and advisor/planner model slots with per-role reasoning effort ([ADR-0033](#)); a Sonnet executor + Opus advisor combination cuts inference cost substantially on routine work (~30–40% by per-token pricing arithmetic; workload-dependent, not a benchmark). Plan mode runs on the planner slot, execution on the executor (§1.3).
 - **Auth resolution** (`runtime/auth.ts`, `auth-config.ts`) — the supported credential is an **API key**: the key configured in Settings (a `0600` file at `~/marvin/auth-config.json`, provider `anthropic` or `openrouter`) wins; otherwise `env ANTHROPIC_API_KEY`. Raw keys are never logged and never returned by any status surface (last-4 hint only). OpenRouter is reached by pointing the CLI at a local proxy (`/api/proxy/openrouter`) that forwards Anthropic-format requests, and every model id is then OpenRouter's ([ADR-0096](#)).
 - **Not a supported credential: a Claude.ai subscription login.** Anthropic's authentication policy ([Legal and compliance](#), February 2026) reserves OAuth sign-in for Claude Code and Anthropic's own apps and requires API keys for products built on the Agent SDK. The older host-credentials detection and the Keychain read for model discovery ([ADR-0029](#)) predate that policy; they are legacy, undocumented here, and not supported.
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8. UI surfaces (macOS SwiftUI app)

The app (`macos/MARVIN/`) is fully native — the Tauri/WebView era was migrated out ship-of-Theseus style ([ADR-0010/0011/0016](#) record the history). Major surfaces:

- **IDE shell** — three panes (file tree / chat / brain-graph), split-view autosave, light-first OKLCH theme with dark mode ([ADR-0006](#)).
- **Chat** — mode + effort controls in the input (`ChatModeToolbar`), per-project persisted tabs, image paste + attachments, plan cards, the two-tier checklist strip, decision sheets, compaction banner ([ADR-0022](#)), scope-met chip strip.
- **Editor** — `STextView` with tree-sitter highlighting (Swift, TS/TSX, JS/JSX, Go, Rust, JSON), diff gutter positioned from real layout fragments, `⌘S` with mtime compare-and-swap.
- **Navigation** — Quick Open (`⌘P`), graph-backed Go-to-Symbol (`⌘T`), ripgrep Find-in-Files with replace-all, file history popover, Quick Look.
- **Terminal** — PTY-backed with ANSI parsing; build-task palette (`⌘⇧B`) discovering tasks from `package.json` / `Makefile` / `Package.swift` / `Cargo.toml`; diagnostics panel parsing compiler output.
- **Source control** — stage/commit/push/pull/fetch through the guarded git channel (§2.5), branch line, remote-error banner with the `stderr` taxonomy.
- **Review window** — §2.3.
- **Backlog panel** — browse / Done / Dismiss / Promote-to-plan (which switches to Plan mode and queues if a turn is live — v0.1.53).
- **Context panel** — the status-bar `ctx` chip opens a live breakdown: exact resident/window % from SDK usage plus per-category estimates (system prompt · tools/MCP · project context · transcript · free).
- **The brain** ([ADR-0019](#)) — a Metal-rendered live state indicator with five behavioral states (idle / thinking / tool / writing / error).

- **Status bar** — health hairline, cost pill, branch badge, model row, graph-ops ratio.

8.1 Health & resilience

`HealthMonitor` demotes to `.offline` only after **3 consecutive** `/api/health` misses (5s timeout, fast re-poll while misses pend) — one slow poll from a busy sidecar no longer tears down and rebuilds the IDE (v0.1.54). One live turn per session is enforced server-side (`POST /api/chat` → 409 rather than evicting; Stop is authoritative via `cancelLiveTurn`); closing the window doesn't kill a running turn (resume via `GET /api/chat/resume`).

9. Background & async

- **Background jobs** (ADR-0038) — `run_background_job({command, reason})` — the tracked replacement for the raw backgrounding mechanisms §2.6 denies (similar name, opposite fate: `run_in_background` is denied, `run_background_job` is the sanctioned tool) — spawns a tracked child, streams an 8KB output tail, and **fires a real follow-up turn on process exit** with the exit code and tail. Limits: ≤ 3 concurrent per session, chain depth ≤ 8 . Shutdown signals (SIGTERM on app quit) are "stopped, not finished" — no spurious failure turns on relaunch (v0.1.48).
 - **Wakeup** (ADR-0031) — `schedule_wakeup` re-invokes MARVIN at a chosen time via a persistent, boot-re-armed scheduler (delay 60s–24h, ≤ 5 pending, re-schedule depth ≤ 3); a fired wakeup **yields** to a live interactive turn and re-arms.
 - **Announce SSE** (ADR-0043) — an always-on per-project stream re-attaches an idle client to any server-initiated turn (job completions, wakeups), with a "background job running" chip.
 - **git-watch** (`sidecar/packages/git-watch/`) — per-workDir commit detection feeding graph auto-rebuild and inline commit surfacing.
 - **The anti-fabrication contract** (§11.5) makes these three mechanisms the *only* honest ways to promise future work.
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10. Distribution & operations

- **Homebrew** (ADR-0023) — `brew tap RobertIlisei/marvin && brew install --cask marvin-ai` (token disambiguated from the unrelated "marvin" cask). The .app bundles the Next.js standalone sidecar + a pinned Node 22 runtime; the SwiftUI process spawns and reaps it, reclaiming port 3030 first (ADR-0035) so "new app on disk, old code in memory" cannot recur.
- **Signing** (ADR-0026) — every release zip carries a minisign (EdDSA) signature; the public key is pinned in the app repo, the tap README, and the cask constant — three copies across two repos, so tampering with one is visibly inconsistent. Cask auto-verify is the ADR's Phase 2.
- **macOS 26 Gatekeeper** (ADR-0027) — ad-hoc-signed bundles are kernel-killed in `/Applications`; MARVIN installs to `~/Applications`, the cask strips quarantine in a `postflight`, and first launch needs the one-time "Open Anyway".
- **Lifecycle** — `bin/marvin start/stop/restart/status/logs/doctor/ knowledge-graph`; `install-macos-app --bundled` builds, bundles, health-probes on a probe port, then installs; `brew uninstall --zap` wipes `~/marvin`.

- **Observability** — `/api/health` reports auth mode + version; the sidecar logs to `~/Library/Logs/MARVIN/sidecar.log`. Optional **Honeycomb OTEL export** (`runtime/honeycomb-telemetry.ts`): per-turn env injection so the SDK-spawned CLI emits spans to the *user's own* Honeycomb account — off unless the user configures it; keys redacted from all logs.
- **CI/release** — tag push → `release.yml` builds the .app on a macOS runner, stamps the version into Info.plist, zips, signs, publishes; the cask is then bumped with the published sha256.

11. Behavioral contracts (`personality.ts`)

The prompt is a contract surface, not vibes. The catalogue (each item a MUST list + MUST-NOT list + narrow fallback test):

1. **Phase gating** — labelled phases, stops between them, no mutation before Phase 6, explicit trivial fast-path.
2. **Definition of Done / match-not-improve** — 3–5 falsifiable bullets before code; Phase 7 verifies those exact bullets; adjacent improvements are surfaced ("noticed in flight, not in scope") and parked or asked — never silently landed. Real-work turns end `**Scope met:**` ... Anything else, or should I stop? plus a machine sentinel for the UI chip strip.
3. **Verify-then-remediate** (v0.1.55) — mechanical failures (typecheck/tests/build) self-remediate ≤3 attempts per milestone with an early stop on identical consecutive errors; MUST NOT claim landed, weaken the DoD, or skip the failing check. Unmet DoD bullets get surface-and-offer: the gap + the one next step, then a gate. A blind retry-until-done loop was deliberately rejected.
4. **Deterministic ADR triggers** — nine MUST-write categories (foundational framework, public API shape, persistent-state schema, security boundary, default-model change, new MCP server, cross-cutting constraint, superseding, user-named) + anti-triggers + the "would future-MARVIN re-derive this in 8 weeks?" test.
5. **Async anti-fabrication** — never narrate a watcher that wasn't armed; the only honest options are block-and-wait, a background job, a wakeup, or handing back.
6. **Post-PR loop** — own the green build: run tests locally, one structured PR comment per run, fix the code under test, cap three run-fix-run cycles, name flakes honestly.
7. **Workflow health** — an injected audit block (Mode A propose / B execute / C defer) catches an in-flight project up on missed phases; a greenfield playbook adapts Phases 2–3 for empty repos.
8. **Personality** — `marvin` (dry wit, always delivers) vs `neutral`, a style layer never a refusal layer; ground-truth facts are placed first in the prompt (the highest-attention slot).

12. API & MCP surface

12.1 HTTP API (sidecar, `localhost:3030`)

Route groups under `sidecar/src/app/api/`: `chat` (+ `announce`, `resume`, `cancel`), `confirm`, `changes`, `files` (`tree` / `content` / `raw` / `status` / `write` / `upload`), `git` (`status` / `diff` / `branch` / `log` / `push` / `pull` / `fetch`), `graph`, `projects`, `sessions`, `skills` (+ `add`), `models`, `auth`, `health`, `context`, `cost`,

`audit`, `backlog`, `terminal`, `honeycomb`, `worktrees`, `practice` (+ `run`, `findings`, `rules`). Details: [docs/reference/api.md](#).

12.2 MCP servers

Server	Type	Tools	Gate posture
<code>marvin-graph</code>	in-process	6 graph tools	allow (read-only)
<code>marvin-memory</code>	in-process	<code>remember</code> , <code>recall</code>	allow (content-class enforced)
<code>marvin-backlog</code>	in-process	<code>backlog_add/list/resolve</code>	allow (content-class enforced)
<code>marvin-control</code>	in-process	wakeups + background jobs	allow (bounded by rails)
<code>playwright</code>	external stdio, opt-in	<code>browser_*</code>	observation auto · interaction confirm · <code>browser_run_code_unsafe</code> deny ; scouts get observation only (ADR-0045)

Browser automation defaults to the Playwright **CLI** (one-shot screenshots, scripted checks, full `playwright test`); the MCP is for stateful `navigate→snapshot→assert` flows, off by default because a browser subprocess per turn is heavy. MCP-vs-CLI selection is itself a deterministic trigger in `personality.ts`.

13. Deliberate non-goals

Explicit product boundaries, each with a recorded reason (`docs/roadmap.md` "Not planned"):

- **No multi-agent implementation** — the founding bet ([ADR-0001](#)); the three read-only subagent carve-outs are not a precedent.
- **No cross-project memory** — projects are isolated workspaces ([ADR-0005](#)); a new project starts from zero, by design.
- **No hosted service / shared state** — local-first is the trust model, not a phase.
- **No cross-platform desktop (yet)** — fully-native macOS is the current commitment; x86_64 and other platforms wait for demand.
- **No auto-model heuristics** — the user routes roles to models explicitly; MARVIN doesn't guess task complexity ([ADR-0002](#)).

14. Building & contributing

This document is the architecture map; the practical loop lives in the repo:

- **Dev setup & build** — `README.md` ▶ *Development setup* (sidecar: `pnpm` + `Next.js dev` on `:3030`; app: `swift build` or `xcodegen` + `Xcode` in `macos/`) and ▶ *Dev loop* for running both together. `bin/marvin install-macos-app --bundled` produces the installable `.app`.

- **Tests** — TypeScript: `npx vitest run` from the repo root (policy, sandbox, registry, and dispatch contracts are the best-covered areas); Swift: `swift test` in `macos/`. Honest scope note: the SDK loop, the UI shells, and most API routes are opportunistically covered at best — `docs/development/testing.md` has the current map.
 - **Conventions** — work lands on `development` and fast-forwards to `main`; a gitleaks pre-commit hook is installed from `scripts/hooks/pre-commit`; CI is `.github/workflows/release.yml` (tag-triggered).
 - **Understanding why** — read `docs/decisions/` (the 51 ADRs) before proposing structural changes, `docs/roadmap.md` for what's in flight, and `CLAUDE.md` for the golden rules any agentic session in this repo is bound by. Material changes should arrive with an ADR of their own.
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Compiled against the 2026-07-02 full feature inventory (51 ADRs, v0.1.55). Corrections welcome — the ADRs are the authoritative record, and this document cites them everywhere it can.